Last Class (Sept 29)

1) Intro to Unsupervised Learning & Problem Statement
2) Clustering & intuition
3) Dunn Index
4) K-Means & its Math Formulation
5) Lloyd's Algorithm
6) K-Means Scratch Implementation
7) Determining K
8) Amazon's customer segmentation with K-Means

Today's class

1) Polls
2) Silhouette Score
3) KMeans: Initialization
4) K-Means ++
5) Limitations of K-Means
6) K-Median and K-Medoid

Case I :

$a = 0.01$
$b = 100$

$\frac{b-a}{b} = \frac{100 - 0.01}{100} \approx 1$

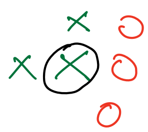
$a < b \Rightarrow b - a > 0$

$$\text{Silhouette score (SS)} = \frac{b-a}{\max(a,b)}$$

$SS \approx 1$

$SS > 0$

Case II :



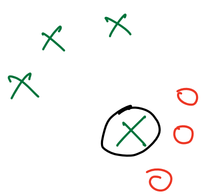
$a \approx b \Rightarrow b - a \approx 0$

$\frac{b-a}{\max(a,b)} \approx 0$

$SS \to 0$

$SS \approx 0$

Case III :



bad/poor/wrong clustering

$a > b \Rightarrow b - a < 0$

$SS < 0$

$a = 100$

$b = 0.01$

$SS \to -1$

$SS \approx -1$ (if $a >> b$)

$$SS = \frac{0.01 - 100}{\max(100, 0.01)}$$

$$= \frac{-99.9}{100} \approx -1$$

Range of SS $\to$ -1 to $+1$

SS (x_i)

SS for entire data = avg of SS for all x_i's